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Rashtreeya Sikshana Samithi Trust

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CircuitFlow

CircuitFlow: Smart Water Management, Drop by Drop

Urban Development, Mobility & Smart Cities: 73

Team Smartopia

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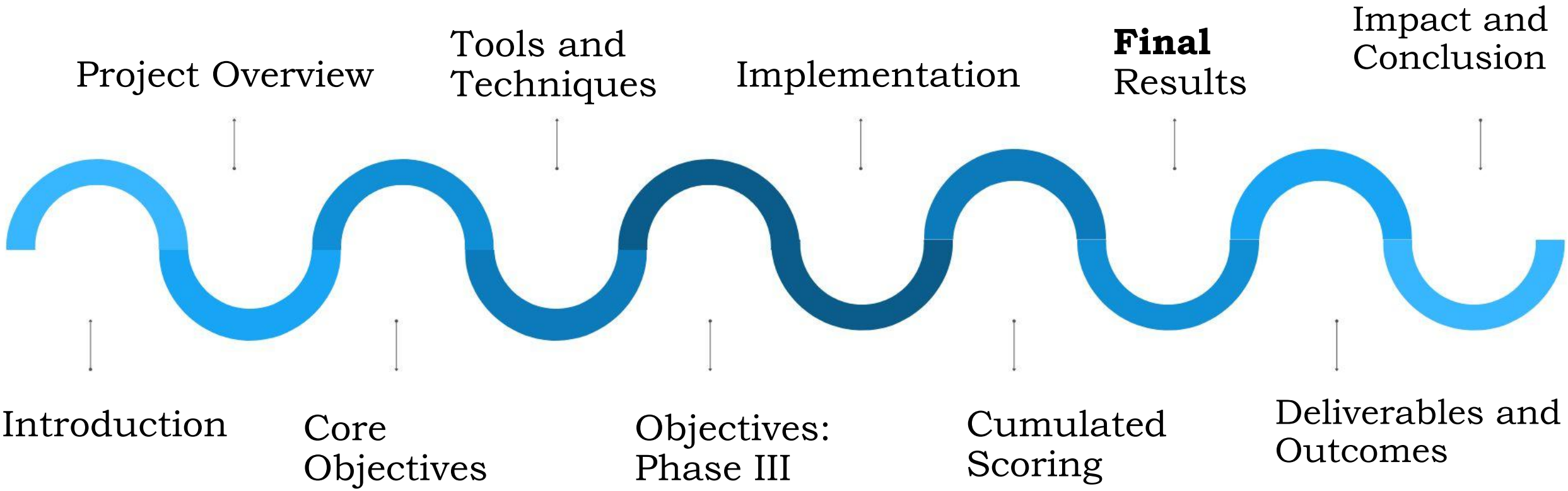


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Outline

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Introduction

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Overview

Smart tap-system for water wastage remediation and *usage analysis* at the household level through real-time data collection and *predictive modelling*.

Importance

- An average Indian household loses nearly 45,500 L through tap-leaks annually, aggravating a global water crisis.
- According to the India water portal report in Bangalore, around 360 million liters of water go waste every day due to leakage and other ‘minor’ reasons.
- This necessitates a structured wastage analysis and remediation system based on leakage-alerting, usage-dashboards, and data-driven regulation.

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Project Overview

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Problem Statement

- Rapid population growth and poor monitoring have intensified domestic water wastage, with Indian households losing about 45,500 L annually.
- Existing systems lack real-time control and user engagement.
- This project develops a smart, sensor-based system with flow-control valves and analytics to detect anomalies, automate corrections, and promote sustainable water use.

Significance

- Rapid population growth and urbanization are straining freshwater resources, while unmonitored household use and unnoticed leaks worsen wastage.
- Smart real-time monitoring can improve awareness, reduce losses, and promote sustainable water consumption in homes.

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Problem Context
• Water scarcity has become a critical global issue, with studies indicating that throughout 2025 nearly 67% of the world's population will experience moderate to high water stress. • In India, urban households consume between 183 and 215 liters of water per person daily, placing immense pressure on already limited resources. • The residential sector accounts for the largest portion of urban water usage, primarily for bathing, washing, flushing, and cooking.Bath and shower accounts for 55% of water-usage in urban households.

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Project Overview

Review of Literature

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Authors	Title of the publication	Foundations of the model	Plausible Upgrades/Fixes
Ruturaj Marathe, Manisha Tapale, Vidya Jadhav, Vrushab Hulbatte, Aishwarya Pawar	**IoT based Water Leakage Detection using Smart Objects for Smart City**	The model used Arduino flow-sensors to perceive leakages, and communicated the leakage data to the PushBullet application.	• Integration with solenoid valves to block leaks • analysis of usage-data and • timed water-flow mechanisms.
Chavhan S., Gawande U. H., Bhoyar D., Rangari S., Gardi M., & Gaikwad A.	**IoT-based Remote Control Water Distribution System (**International Journal of Intelligent Systems and Applications in Engineering)	• The model enables the control of water valves remotely through a mobile interface. • The system used a microcontroller(ESP8266), relay drivers, and solenoid valves to control water flow, aiming to reduce water losses and increase response time in saving water in large distribution pipelines.	• It relied on manual control rather than automatic shut off when water is left running. • It is aimed to be handled by the skilled personnel such as water operators, not individuals at the household level. • Integration of daily usage tracking and AI for analysing usage patterns.

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Project Overview

Review of Literature

CircuitFlow

Authors	Title of the publication	Foundations of the model	Plausible Upgrades/Fixes
Vatsala Sharma, Kamal Nayanam, Himani	**Arduino based Smart Water Management**	• The paper lays an excellent foundation for IoT water management with the usage of sensors, though it focuses on automatic water irrigation. • The system uses Arduino (NodeMCU ESP8266), ultrasonic sensors to detect extent of water flow and Blynk app integration.	• Automatic shutoff which reduces wastage. • potential for AI-driven usage optimization • ***local data logging***, making it not entirely reliant on stable internet connection.
Isuru Sachitha Herath	**IoT based Intelligent Domestic Water Management system**	• The paper proposes “Smart Water Buddy”, an IoT and machine learning-based system designed to monitor, analyze, and optimize domestic water usage.	• There is no failsafe mechanism to evade data loss/inconsistencies due to loss of internet connectivity.

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Core Objectives

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Main Objective
• To effectively identify and address unattended water loss in the households of major Indian cities.
Specific Objectives
• To develop a model that detects and remediates leaking and overflowing taps. • To demonstrate a dashboard on users' water utilization.
Expected Contributions
• The model proposes to add a data-driven, adaptive approach to existing models on the subject.

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Tools and Techniques

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Hardware
Arduino UNO (Microcontroller Unit)
YF-S201 Water Flow Sensor
HW201 IR Obstacle Sensor
LEDs (Red & Green)
Buzzer
Jumper Wires, Resistors, Power Supply
Water Flow Valve

Software
Flask
Pyserial
Multithreading
Scheduling
Web development tools (HTML, CSS, JavaScript)
Chart.js
Python
Android WebView

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Objectives

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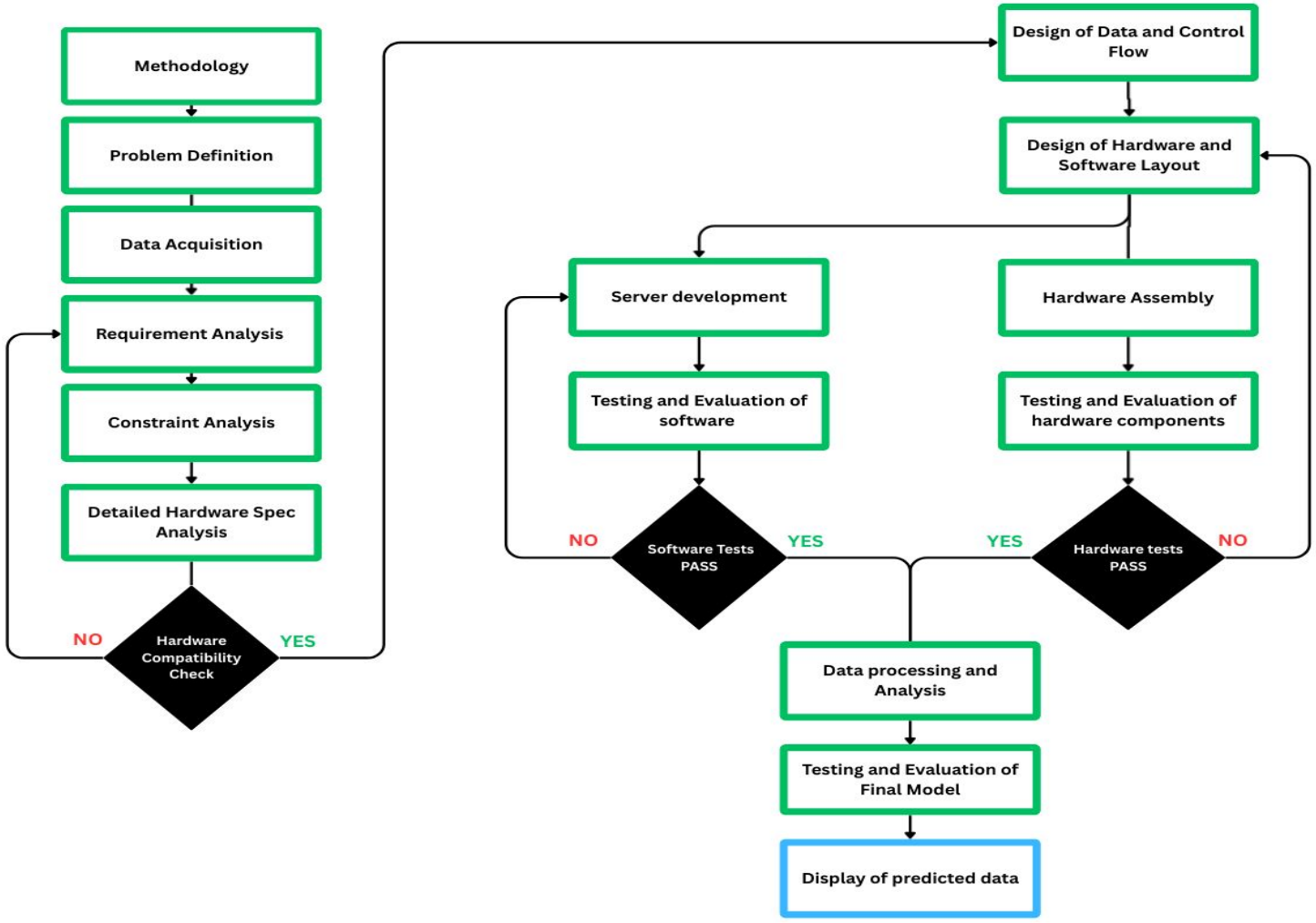
PHASE - III

- Accurately display *flow data* from the *water flow sensor* for fixed intervals of time, in order to understand variations in flow.
- To achieve full hardware-software integration.
- Integration of the preventive valve.
- To log daily water consumption values.
- To predict and display the approximate usage of water for the next day, based on previous data.
- Visual and Auditory feedback.
- To develop a remote-access mechanism for viewing live data from smartphones.

Project Implementation

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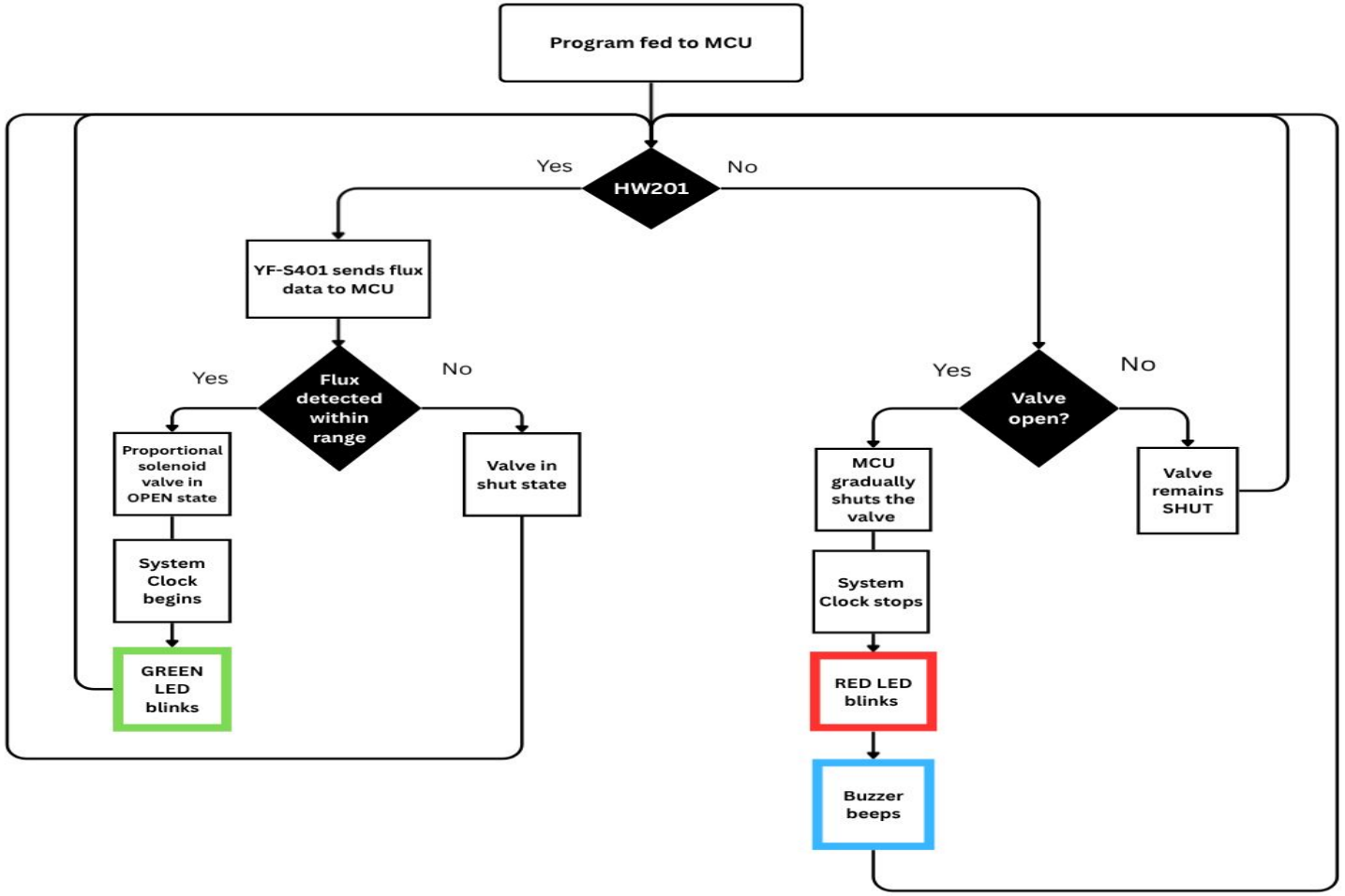
Plan of Action



Legend	
Methodology	Accomplished task

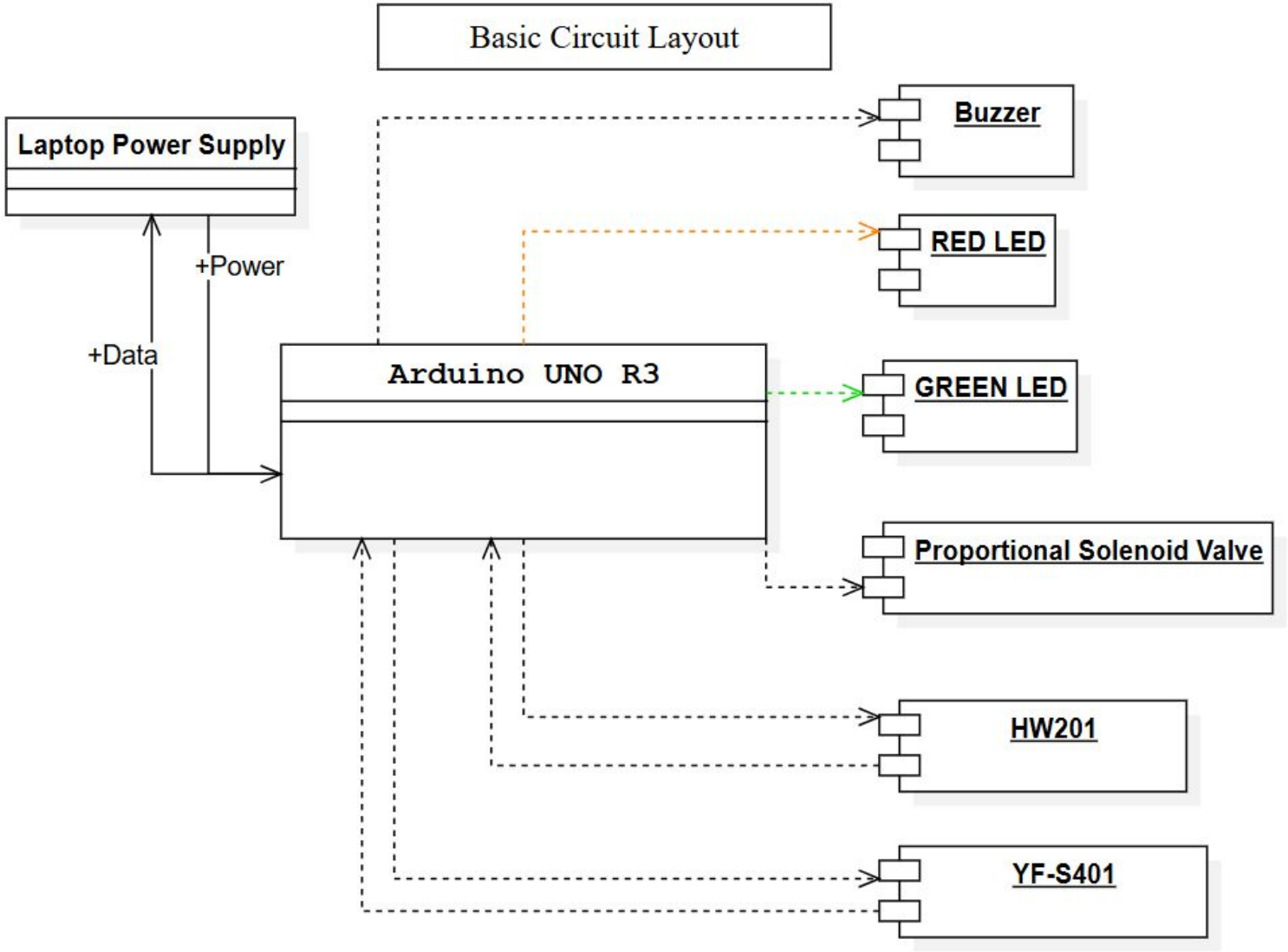
Project Implementation

System Architecture



Project Implementation

Hardware Architecture



Review of the Constraints

- For **maximum power efficiency**, the model can be operated in a specified timeframe (8 to 10 AM) , during which most indian households have *maximum consumption*.
- The model will operate only for a **maximum water flow pressure of 0.8MPa**.
- The model will operate on an **adjusted temperature range of 6-45℃** which is the usual water conditions.

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Internal Scoring Scheme

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Developed by Smartopia

Description	Score
Hardware	
Hardware component fails any one test	0
Hardware component passes tests, but without satisfactory results	0.5
Hardware passes all tests satisfactorily.	1
Software	
Software successfully captures and stores **all** serial Arduino data.	1
Software fails to capture **all** serial Arduino data.	0

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Cumulated Hardware Scoring

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Attribute	Assigned Score	Comments
Arduino UNO R3	1	Integration with other components is seamless and reliable, low response time.
YF-S201 Flow Sensor	0.5	The sensor detects the change in flow rate, low response time, low accuracy in flow-values.
HW - 201 IR Sensor	0.5	Detection range is constricted to 30cm, fast response time.
LEDs - Red and Green	1	The LED lights up at its maximum intensity.
Arduino Buzzer	0.5	The buzzer generates a sharp tone, but relatively low response time.
Valve	0.5	The valve shuts off on receiving signals, but relatively high response time.

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Cumulated Software Scoring

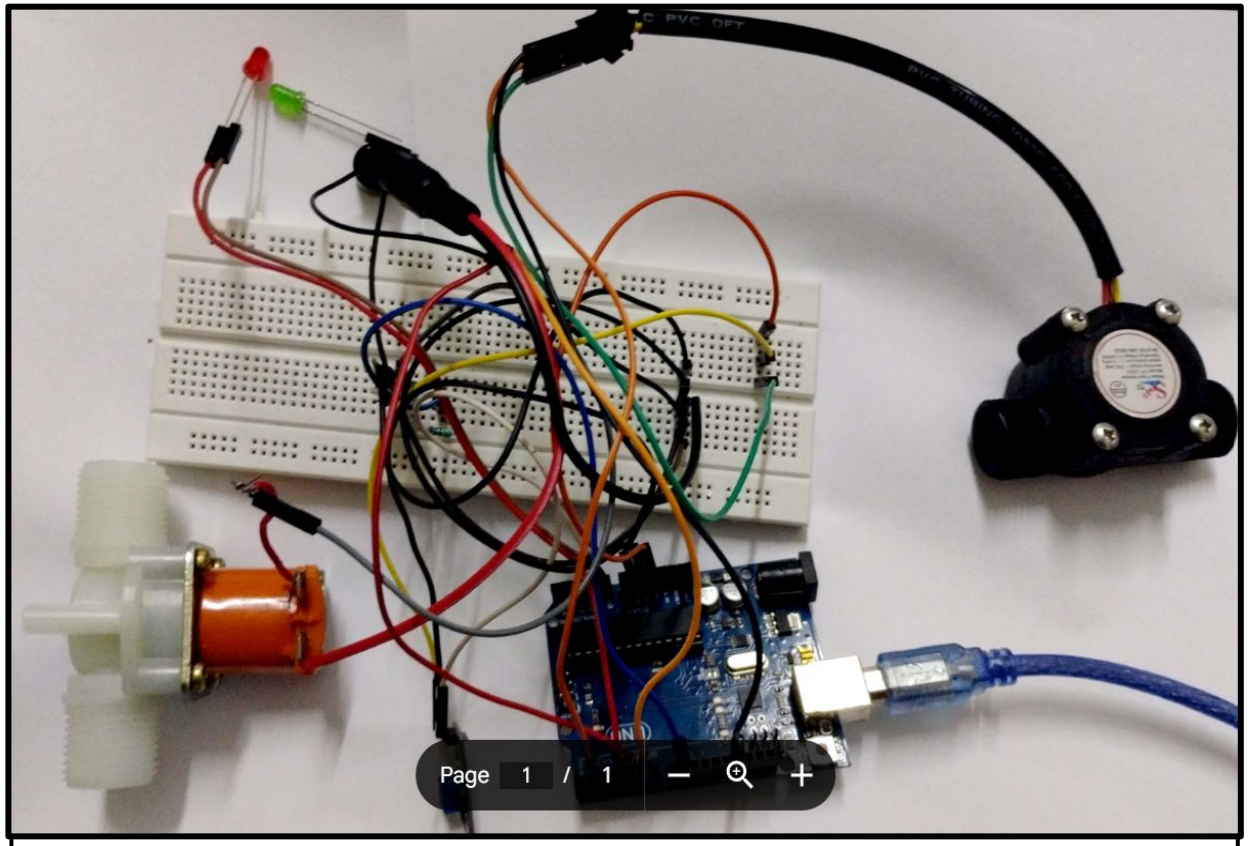
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Attribute	Assigned Score	Comments
Flask Server	0.5	Flask sends data accurately and calls functions as per the scheduled time, but can face lags during startup.
Predictive Algorithm	1	The algorithm is designed for approximate calculations, and returns the same.
Serial Data Recognition	1	Serial data is received without data loss.
Threading	1	Threads run smoothly and without significant lags.
Android WebView	1	Android application is lightweight and has low latency. It facilitates remote access.
Interactive User Interface	1	The User-Interface features user-friendly navigation and informatics. The UI updates as per the predicated schedule.
Arduino	1	All components respond to the code with low latency.

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Final Results

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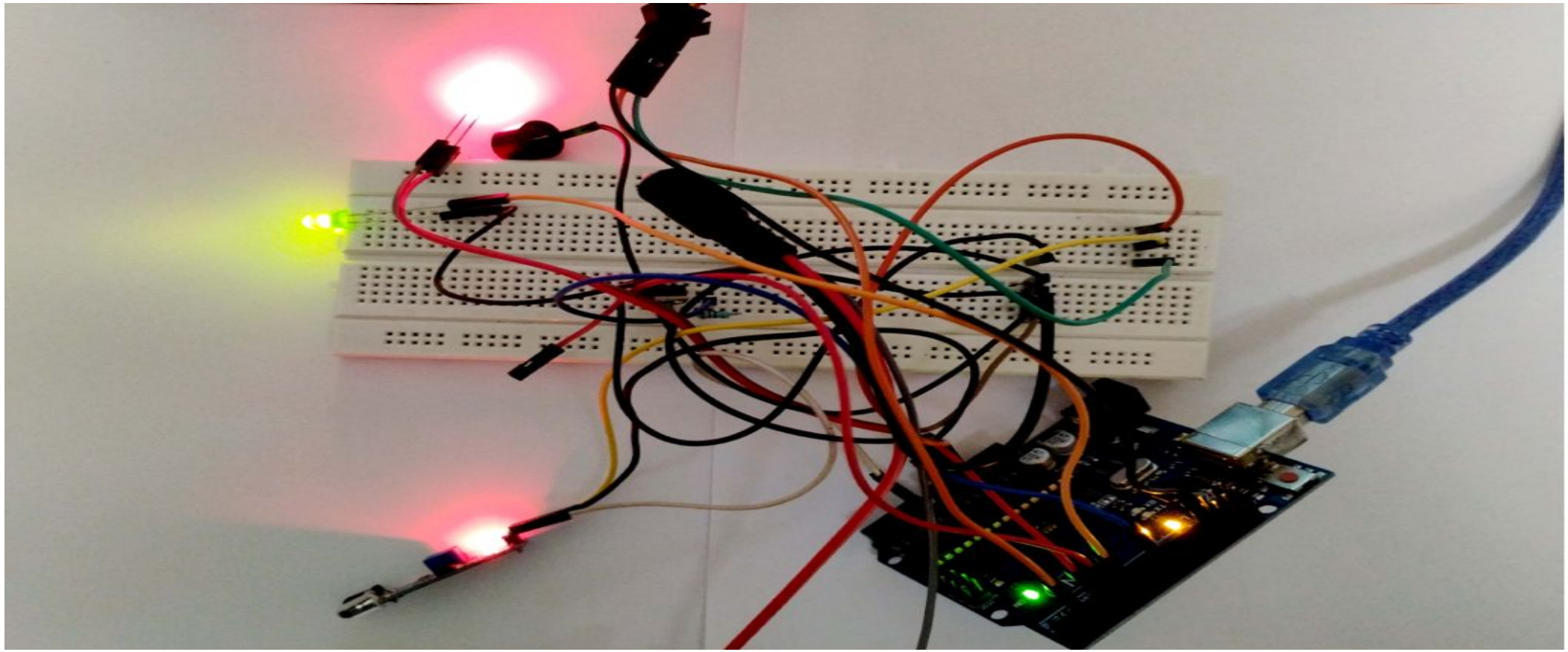


A view of the hardware assembly

Assembled Components
YF-S201
Arduino - UNO R3
Red and Green LED
HW-201
Proportional Solenoid Valve
Buzzer
Water Flow Valve

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Final Results

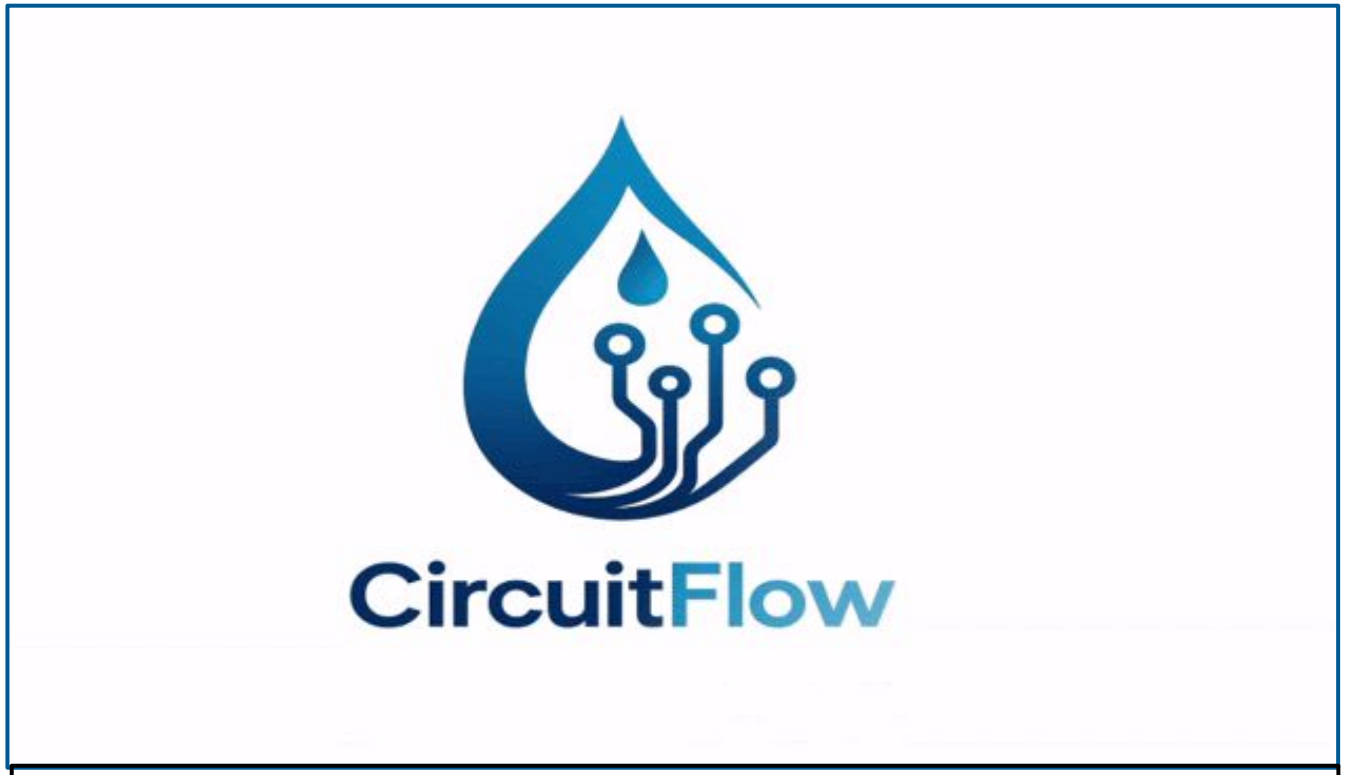


System in operation

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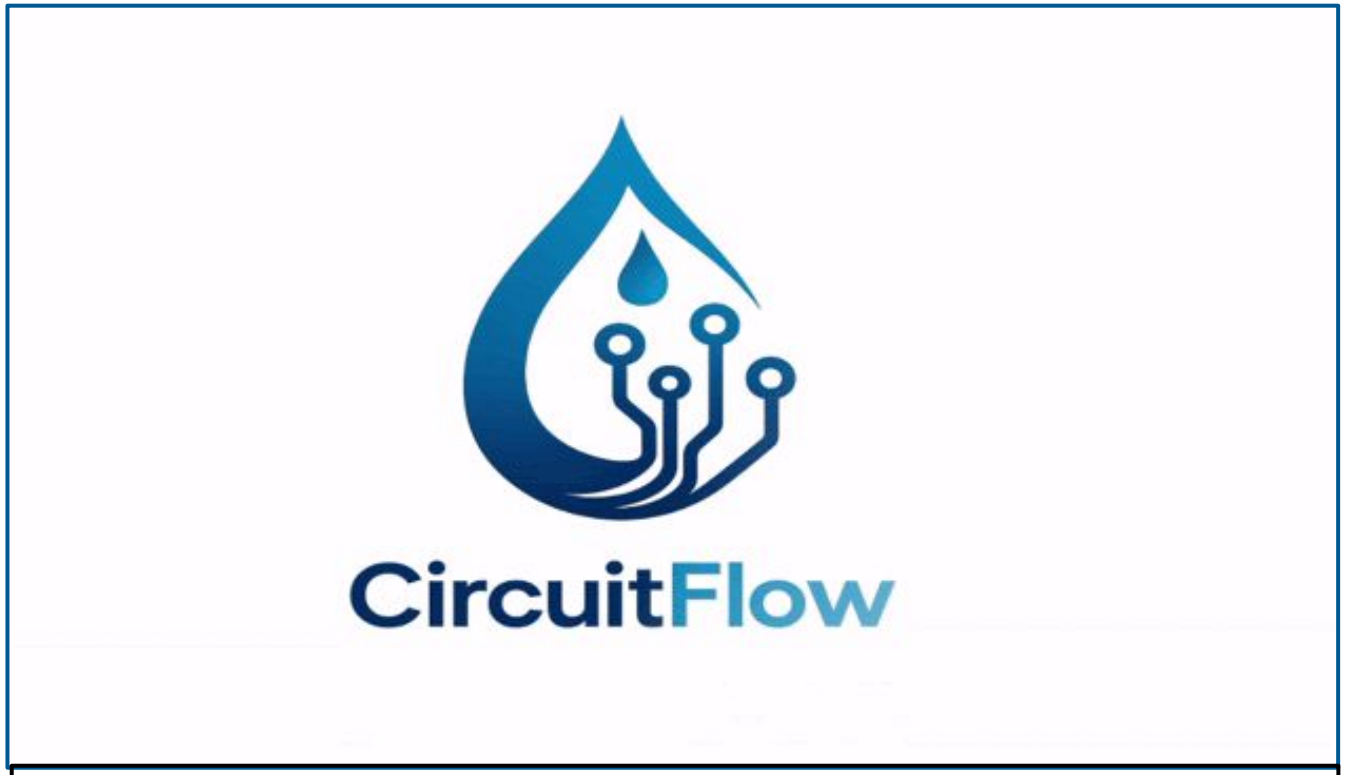
Live WebViewing through the CircuitFlow Android application

The Perception Layer

The Interactive User Interface in action

- A flask server was created to host the basic web pages, which were designed using HTML, CSS and JavaScript.
- The flask app was accessed through browser and Android WebView on the application designed by Smartopia.

Final Results



Live online interface showcasing the sensor data during trial runs

The Perception Layer

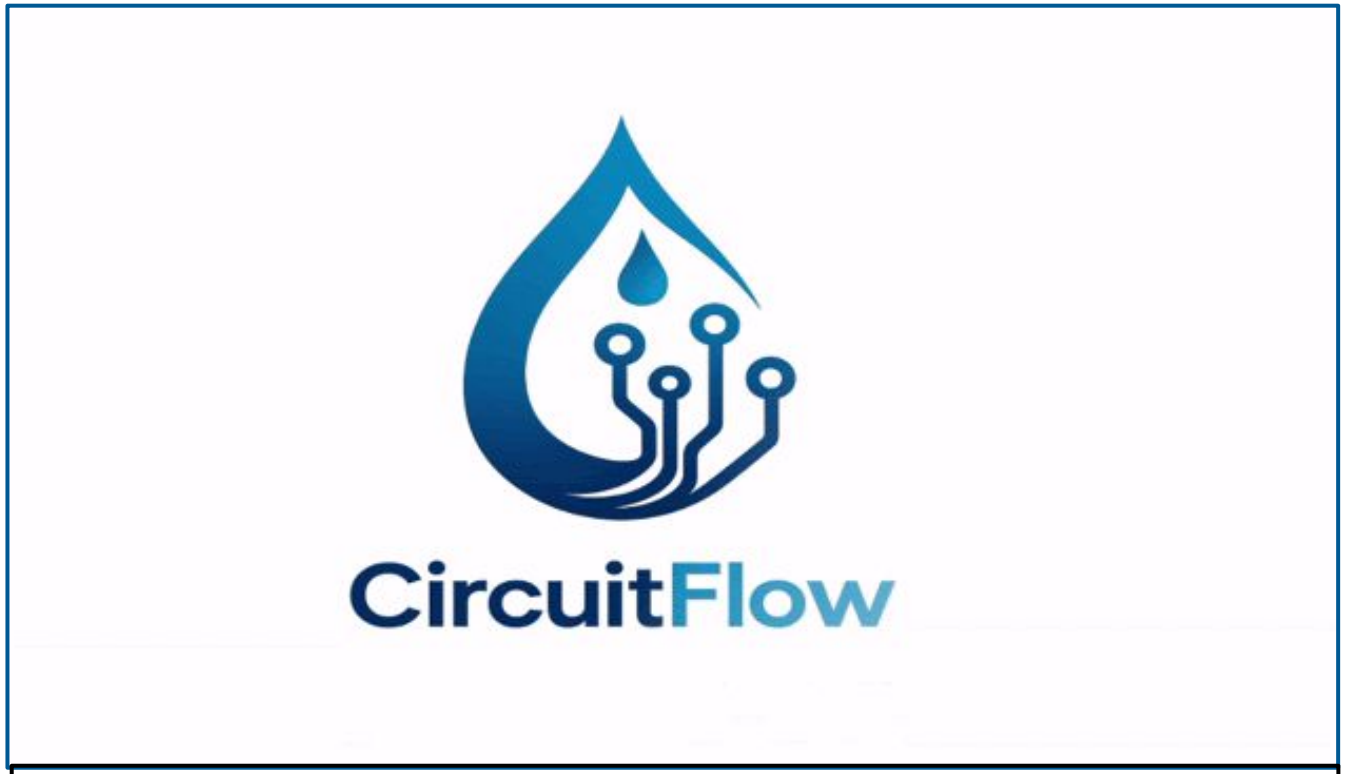
The live data reader

- The live chart receives data pipelined from Arduino through pyserial, which allows python to read serial values from ports.

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Final Results

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Live predictions as made by CircuitFlow

The Analysis Layer

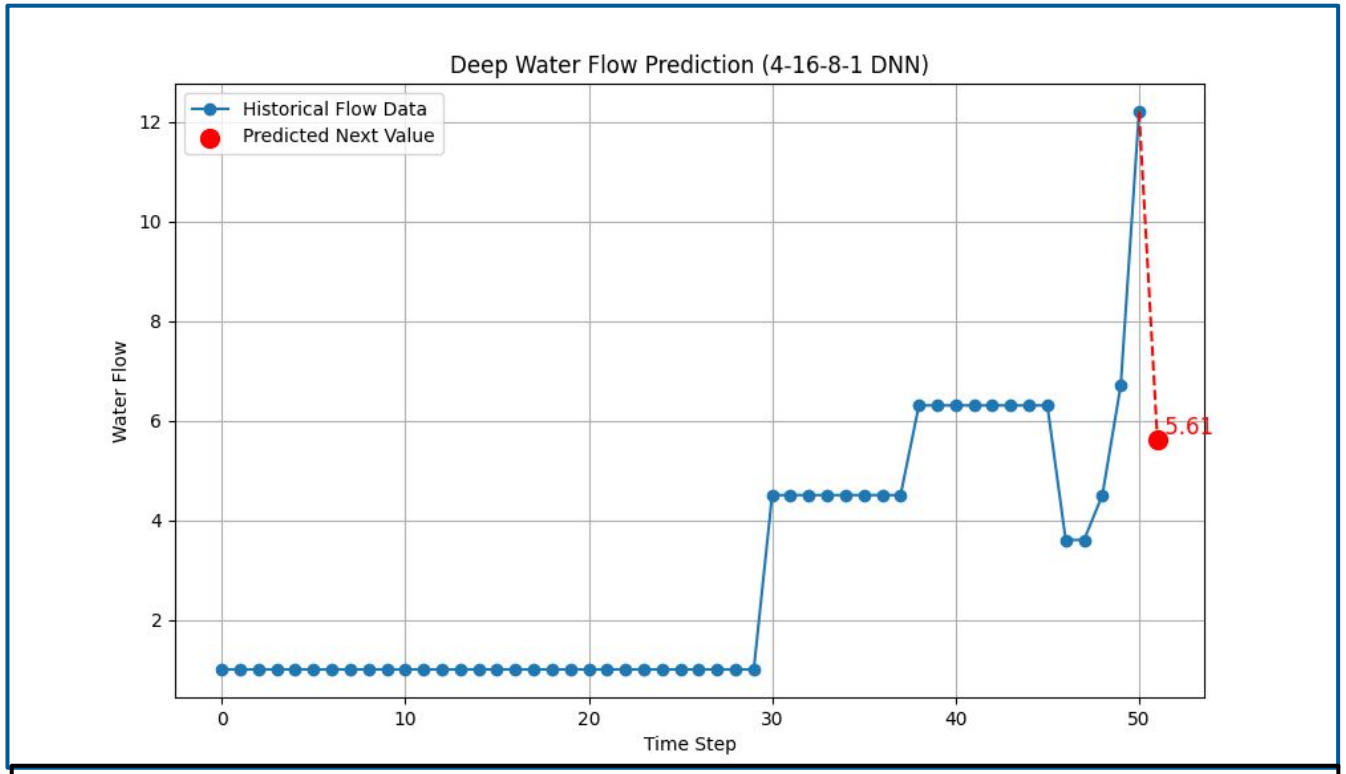
Live Predictions

- The animation showcases live predictions as made by the Circuit Flow model, when trained upon sample usage data.

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Final Results

CircuitFlow



Live predictions as made by CircuitFlow

The Analysis Layer

Live Predictions

- The image showcases live predictions as made by the Circuit Flow model, when trained upon sample usage data.
- A sample plot was created for representation.

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Final Results

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The Analysis Layer

Live data logging

- The program is scheduled to log the daily volume of water to a local file, at 10 AM everyday.
- This enables the prediction algorithm to read the daily water-use data points, and generate a valid prediction based the user's previous usage habits.

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Final Results

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Flow of sensor data from Arduino

Actuation Layer

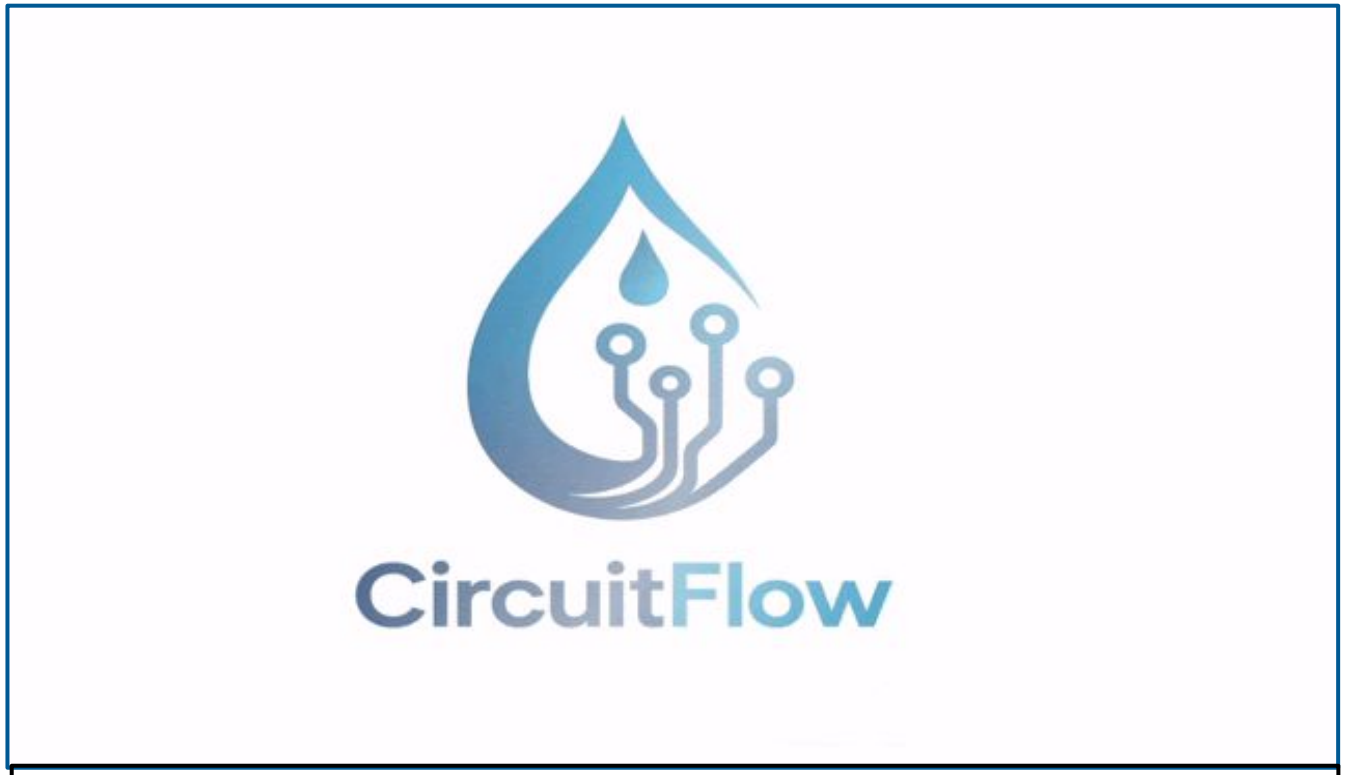
Sending sensor data from Arduino

The animation depicts the passing of serial data from Arduino, which was achieved by team Smartopia.

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Final Results

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Extraction of flow-data

Actuation Layer

Behavioral Signal Extraction

The animation depicts live sensor data retrieval from Arduino UNO.

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Deliverables and Outcomes

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Summary of Key Deliverables of Phase - III
Full hardware-software integration.
Live display of sensor data.
Server deployment
Predictive analysis of data
Remote Access through WebView

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Proposed Impact

Final Results

- Systematic analysis of water-consumption data of households in major Indian cities.

Potential Impact

- For every liter of water saved every day, the annual savings aggregate to over 1000 olympic pools.
- This model could significantly assist in combating the existing water-exigency.

Future Research

This project enables research in IoT-based systems, training of machine-learning models based on household water consumption.

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Conclusion

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Summary:

By integrating low-power sensors, a flow-control valve(actuator), and a local real-time analytical dashboard, the system promotes sustainable usage behaviour.

Further Work:

Further extensions to the proposed model include mobile-based monitoring, and real-time regulations for extended time frames.

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Proposed Roadmap

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Research Proposal

This project holds potential for research-extension.

Team Smartopia seeks to extend **CircuitFlow** further through:

- More rigorous predictive analysis of data.
- Prediction-based early warning for excessive usage of water

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References

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List of References:

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- BelTechno Global:Water Leakage, A Big Issue In Water Distribution And Storage System In India.
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- Smart System for Stabilizing Water Flow Output on Android- Based Taps. Iwan Suhardi et al.(2023).
- Chavhan, S. , Gawande, U.H. , Bhoyar, D. , Rangari, S. , Gardi, M. and Gaikwad, A. (2023). IoT-based Remote Control Water Distribution System. International Journal of Intelligent Systems and Applications in Engineering. 12, 7s (Dec. 2023), 572–583.
- Vatsala Sharma , Kamal Nayanam , Himani, 2020, Arduino based Smart Water Management, INTERNATIONAL JOURNAL OF ENGINEERING RESEARCH & TECHNOLOGY (IJERT) Volume 09, Issue 08 (August 2020)

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